Semantic Exploration Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Analysis of Lag-Based Semantic Distance

PENPAL Project

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Contents

1 Overview

The semantic exploration outputs now use the **current lag-based structure**:

- `k`: lag
- `distance`: mean semantic distance for that lag
- `agent`: `"interleaved"` for the full exchange sequence, or `author_1` / `author_2` for speaker-specific trajectories

The processed filename is still `semantic_exploration_binned.parquet`, but the current generator computes **lag-based distances**, not the older non-overlapping window-centroid bins.

This notebook therefore no longer assumes old turn-level exploration tables. Instead it analyses:

1. **Interleaved sequence exploration** across conditions
2. **Speaker-specific exploration** using recovered role metadata

Important comparison note:

- **Slot 1 / Slot 2** is the directly comparable cross-condition axis.
- **Human / AI** in Human-AI and **Starter-side / Non-starter-side** in same-type conditions are only used as within-condition descriptive views.

Primary cross-condition object. Per story, we compute the trapezoidal area under each slot's mean distance(k) curve over the intersection of k values present for both slots, and take the unsigned difference:

$$A_{\text{semexp}}(i) \ = \ \left|\, \text{AUC}_{\text{slot}=1,\,i} - \text{AUC}_{\text{slot}=2,\,i} \,\right|.$$

This is label-invariant and comparable across conditions. The lag-binned LMM slot contrast and the log-distance slope view are retained as supplemental. Starter-mechanism and Human-AI speaker-type decompositions are deferred.

Article-use note. The primary claim should come from the per-story AUC asymmetry test. The interleaved-distance model answers a different question: whether the whole story sequence explores semantic space differently by condition, not whether the two contributors are more asymmetric.

2 Load Data

Table 1: Loaded semantic exploration rows by condition and agent view

condition	agent	n_stories	n_rows	min_k	max_k
Human-AI	author_1	97	633	2	8
Human-AI	author_2	97	633	2	8
Human-AI	interleaved	97	873	2	10
Human-Human	author_1	36	252	2	8
Human-Human	author_2	36	252	2	8
Human-Human	interleaved	36	324	2	10
AI-AI	author_1	40	280	2	8
AI-AI	author_2	40	280	2	8
AI-AI	interleaved	40	360	2	10

3 Preprocessing

```
## Interleaved rows: 1557
```

```
## Speaker rows: 2330
```

4 Primary: Per-Story AUC Asymmetry

Table 2: Stories with per-story AUC computable in both slots

condition	n_stories
Human-AI	97
Human-Human	36
AI-AI	40

Table 3: Per-story |AUC(slot 1) - AUC(slot 2)| by condition

condition	n	mean_A	sd_A	median_A
Human-AI	97	0.201	0.150	0.180
Human-Human	36	0.192	0.152	0.156
AI-AI	40	0.168	0.143	0.127

```
## One-way ANOVA: A_semexp ~ condition
##              Df Sum Sq Mean Sq F value Pr(>F)
## condition     2  0.032 0.01586   0.716   0.49
## Residuals   170  3.764 0.02214
##
## Kruskal-Wallis (robust): A_semexp ~ condition
##
##  Kruskal-Wallis rank sum test
##
## data:  A_semexp by condition
## Kruskal-Wallis chi-squared = 1.8926, df = 2, p-value = 0.3882
##
##
## Pairwise Wilcoxon (BH-adjusted):
##
##  Pairwise comparisons using Wilcoxon rank sum test with continuity correction
##
## data:  df_auc_stories$A_semexp and df_auc_stories$condition
##
##             Human-AI Human-Human
## Human-Human 0.66     -
## AI-AI       0.52     0.66
##
## P value adjustment method: BH
```

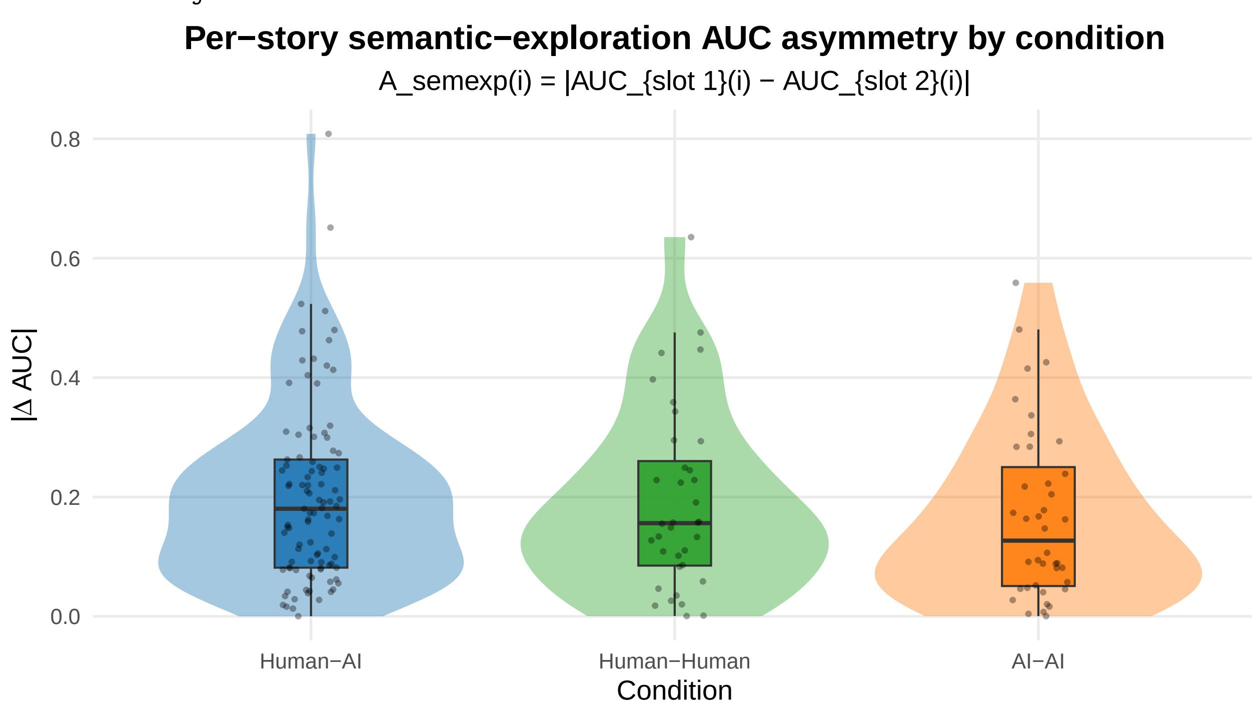


Table 4: Secondary: per-story |log(distance) ~ log(k) slope diff| by condition

condition	n	mean_A	median_A
Human-AI	97	0.146	0.133
Human-Human	36	0.161	0.136
AI-AI	40	0.112	0.088

5 Bootstrap CIs and Effect Sizes on `A_semexp`

Table 5: Bootstrap 95% CI on per-condition mean of A_semexp (R = 10,000)

condition	n	mean	ci_lower	ci_upper
Human-AI	97	0.201	0.172	0.231
Human-Human	36	0.192	0.145	0.243
AI-AI	40	0.168	0.126	0.212

Table 6: Pairwise Cliff's delta on A_semexp

contrast	n_a	n_b	cliff_delta	magnitude
Human-AI vs Human-Human	97	36	0.050	negligible
Human-AI vs AI-AI	97	40	0.149	small
Human-Human vs AI-AI	36	40	0.101	negligible

6 Planned Contrast: HA vs (HH + AA)/2

Table 7: Planned contrast: A_semexp in HA vs the average of HH and AA

contrast	estimate	se	t	df	p	ci_lower	ci_upper
Human-AI vs (Human-Human + AI-AI)/2	0.021	0.023	0.931	157.412	0.354	-0.024	0.066

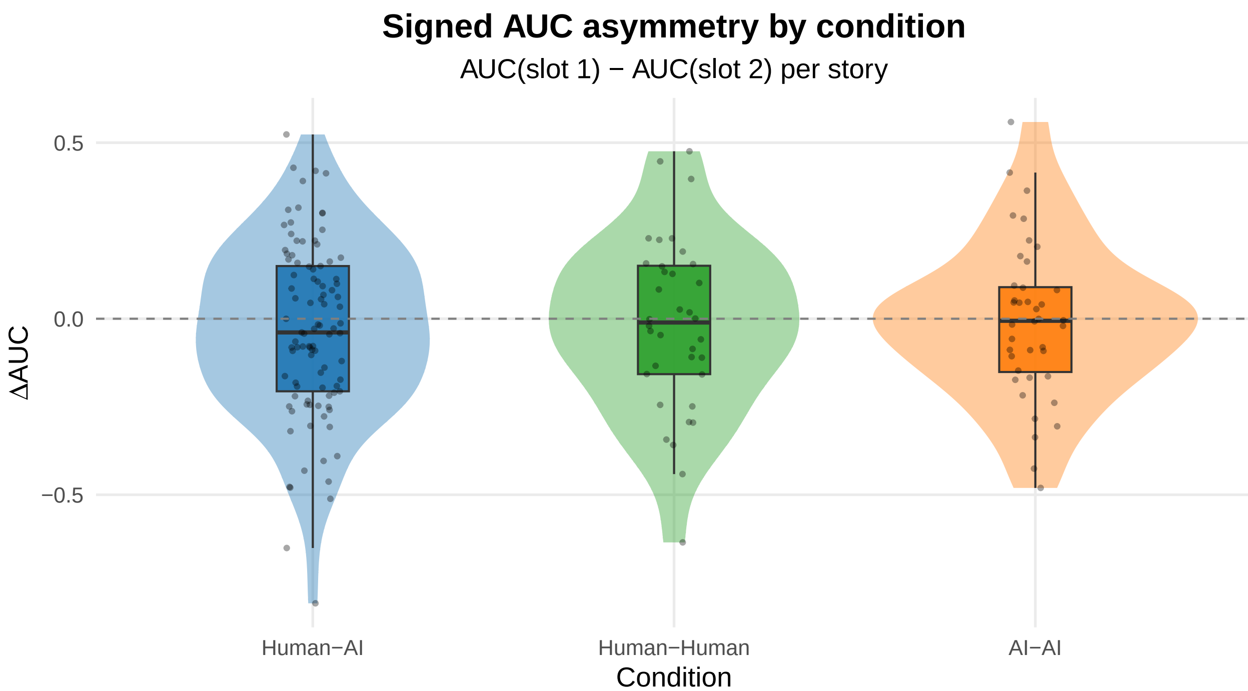
7 Directional Stability of Signed Semantic-Exploration AUC Asymmetry

Tests whether the *direction* of the slot-1 vs slot-2 AUC difference is consistent across stories within each condition. The signed quantity is $\text{signed}_i = \text{AUC}_{\text{slot }1,i} - \text{AUC}_{\text{slot }2,i}$.

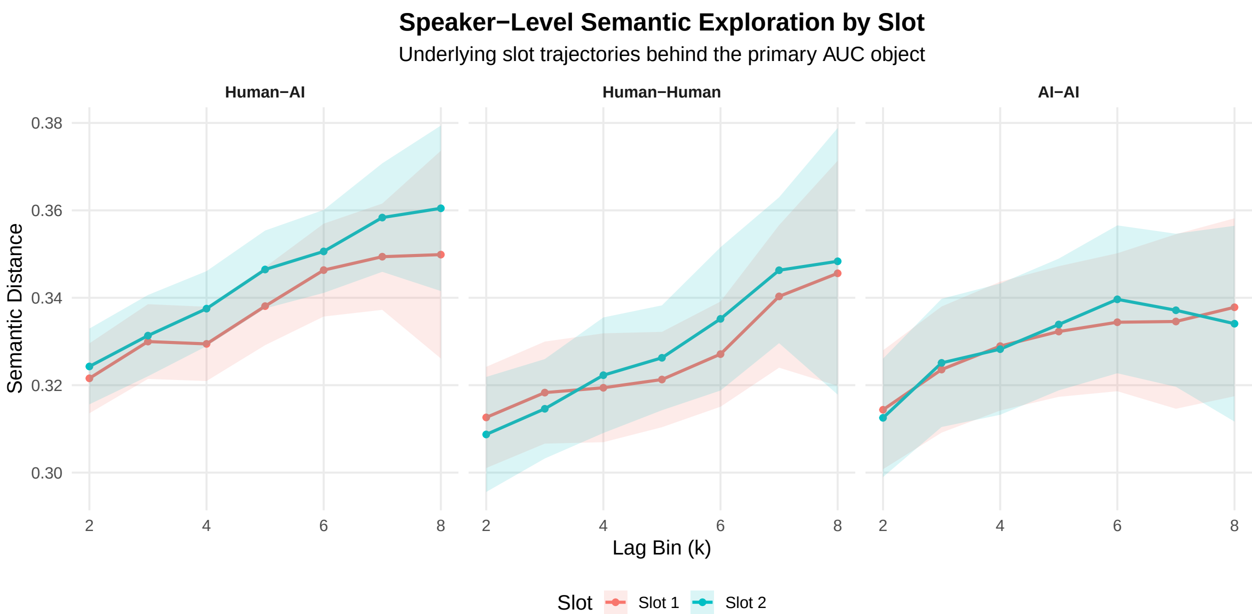
Table 8: Per-condition directional stability of signed AUC asymmetry

condition	n	mean	sd	stability	sign_consistency	one_sample_t_p
Human-AI	97	-0.033	0.249	0.133	0.567	0.193
Human-Human	36	-0.018	0.247	0.071	0.528	0.671
AI-AI	40	-0.007	0.222	0.034	0.550	0.833

```
## Brown-Forsythe variance-equality p = 0.5360
```



8 Supplemental: Speaker-Level Slot Trajectories

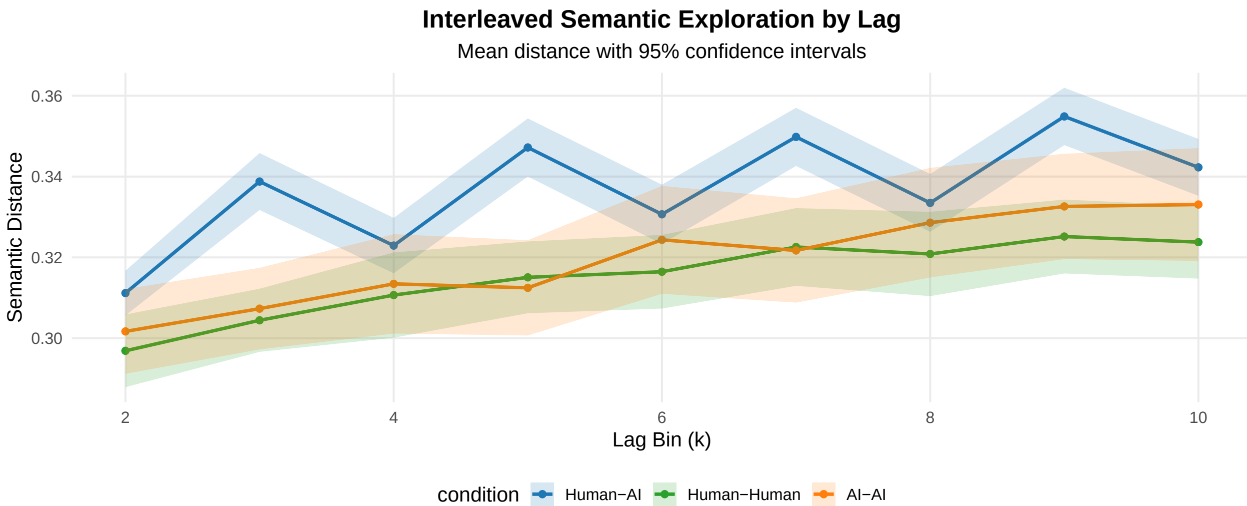


9 Supplemental: General Condition-Level Comparison

Table 9: Interleaved semantic distance by lag and condition

condition	k	mean_distance	sd_distance	n
Human-AI	2	0.311	0.028	97
Human-AI	3	0.339	0.035	97
Human-AI	4	0.323	0.034	97
Human-AI	5	0.347	0.036	97
Human-AI	6	0.331	0.037	97
Human-AI	7	0.350	0.036	97
Human-AI	8	0.333	0.036	97
Human-AI	9	0.355	0.036	97
Human-AI	10	0.342	0.035	97
Human-Human	2	0.297	0.028	36
Human-Human	3	0.304	0.024	36
Human-Human	4	0.311	0.032	36
Human-Human	5	0.315	0.027	36
Human-Human	6	0.316	0.028	36
Human-Human	7	0.323	0.029	36
Human-Human	8	0.321	0.032	36
Human-Human	9	0.325	0.028	36
Human-Human	10	0.324	0.028	36
AI-AI	2	0.302	0.034	40
AI-AI	3	0.307	0.033	40
AI-AI	4	0.313	0.040	40
AI-AI	5	0.312	0.038	40
AI-AI	6	0.324	0.043	40
AI-AI	7	0.322	0.042	40
AI-AI	8	0.329	0.044	40

condition	k	mean_distance	sd_distance	n
AI-AI	9	0.333	0.042	40
AI-AI	10	0.333	0.045	40




```
## Interleaved model summary:

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: distance ~ k * condition + (1 | conversation_id)
##    Data: df_interleaved
##
## REML criterion at convergence: -7579.6
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.5679 -0.5969  0.0001  0.6041  3.2051
##
## Random effects:
##  Groups          Name        Variance  Std.Dev.
##  conversation_id (Intercept) 0.0009884 0.03144
##  Residual                    0.0002948 0.01717
## Number of obs: 1557, groups:  conversation_id, 173
##
## Fixed effects:
##                          Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)             3.171e-01  3.514e-03  2.331e+02  90.242  < 2e-16 ***
## k                       3.275e-03  2.251e-04  1.381e+03  14.550  < 2e-16 ***
## conditionHuman-Human   -2.181e-02  6.755e-03  2.331e+02  -3.228 0.001425 **
## conditionAI-AI         -2.177e-02  6.504e-03  2.331e+02  -3.347 0.000952 ***
## k:conditionHuman-Human  1.828e-05  4.326e-04  1.381e+03   0.042 0.966306
## k:conditionAI-AI        7.413e-04  4.165e-04  1.381e+03   1.780 0.075333 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) k      cndH-H cAI-AI k:cH-H
## k           -0.384
```

```
## cndtnHmn-Hm -0.520  0.200
## condtnAI-AI -0.540  0.208  0.281
## k:cndtnHm-H  0.200 -0.520 -0.384 -0.108
## k:cndtAI-AI  0.208 -0.540 -0.108 -0.384  0.281
```

```
##
## Condition contrasts (interleaved view):
```

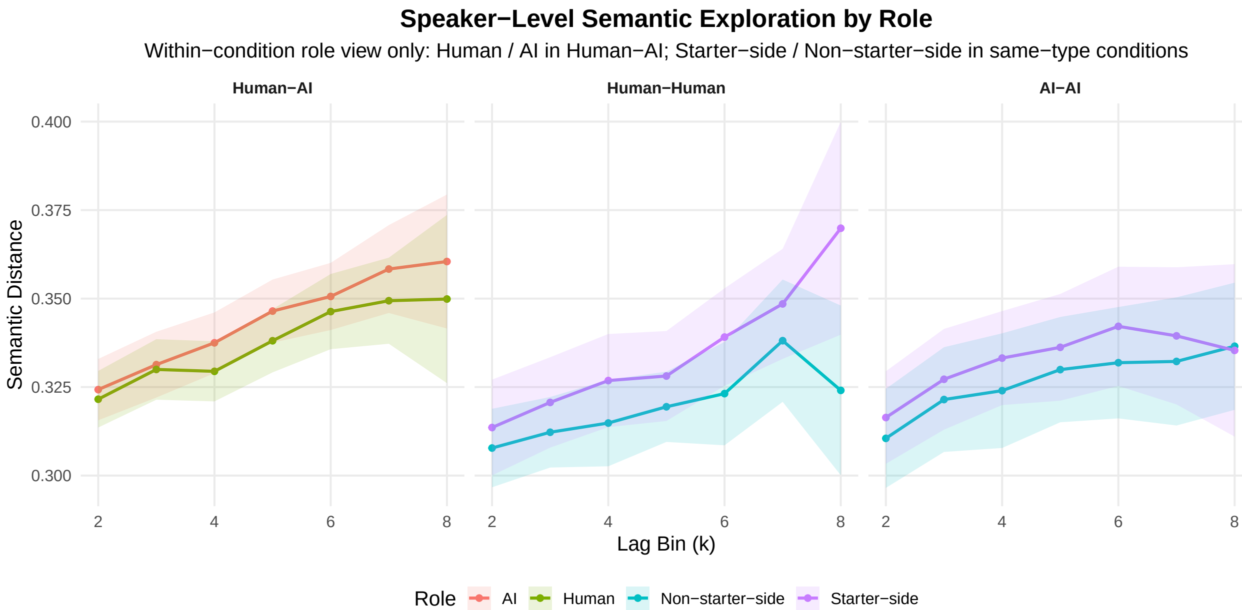
```
##  contrast                    estimate      SE  df t.ratio p.value
##  (Human-AI) - (Human-Human)  0.02170 0.00624 170   3.479  0.0019
##  (Human-AI) - (AI-AI)        0.01732 0.00600 170   2.885  0.0133
##  (Human-Human) - (AI-AI)    -0.00438 0.00734 170  -0.596  1.0000
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests
```

10 Supplemental: Role-Level View

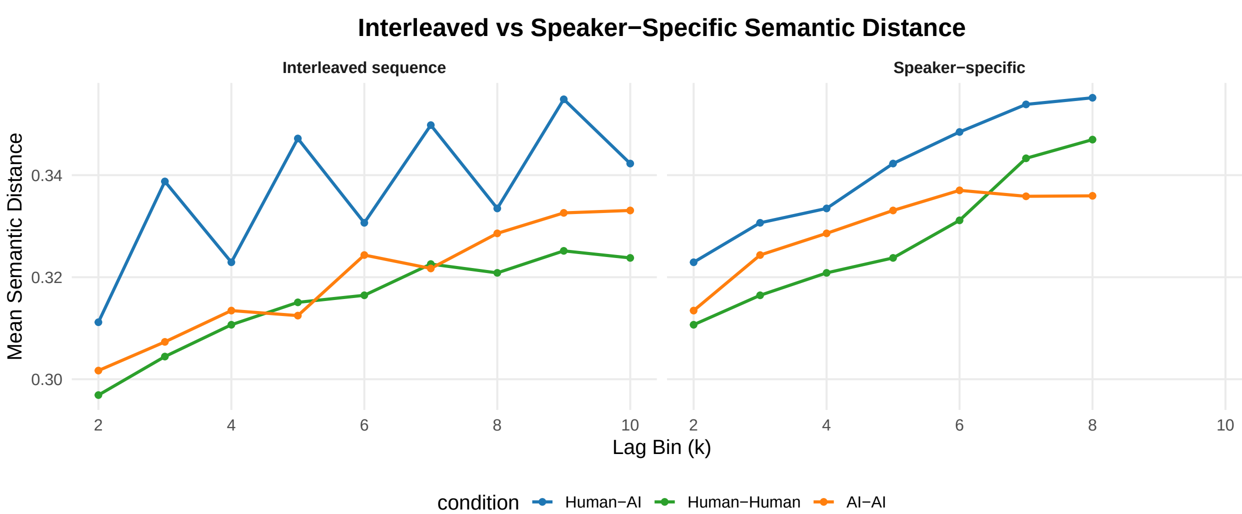
Table 10: Speaker-level semantic distance by condition, role, and lag

condition	condition_role	k	mean_distance	n
Human-AI	AI	2	0.324	97
Human-AI	AI	3	0.331	97
Human-AI	AI	4	0.338	97
Human-AI	AI	5	0.346	97
Human-AI	AI	6	0.351	97
Human-AI	AI	7	0.358	96
Human-AI	AI	8	0.360	52
Human-AI	Human	2	0.322	97
Human-AI	Human	3	0.330	97
Human-AI	Human	4	0.329	97
Human-AI	Human	5	0.338	97
Human-AI	Human	6	0.346	97
Human-AI	Human	7	0.349	96
Human-AI	Human	8	0.350	52
Human-Human	Non-starter-side	2	0.308	36
Human-Human	Non-starter-side	3	0.312	36
Human-Human	Non-starter-side	4	0.315	36
Human-Human	Non-starter-side	5	0.319	36
Human-Human	Non-starter-side	6	0.323	36
Human-Human	Non-starter-side	7	0.338	36
Human-Human	Non-starter-side	8	0.324	36
Human-Human	Starter-side	2	0.314	36
Human-Human	Starter-side	3	0.321	36
Human-Human	Starter-side	4	0.327	36
Human-Human	Starter-side	5	0.328	36
Human-Human	Starter-side	6	0.339	36
Human-Human	Starter-side	7	0.349	36
Human-Human	Starter-side	8	0.370	36
AI-AI	Non-starter-side	2	0.311	40
AI-AI	Non-starter-side	3	0.321	40
AI-AI	Non-starter-side	4	0.324	40

condition	condition_role	k	mean_distance	n
AI-AI	Non-starter-side	5	0.330	40
AI-AI	Non-starter-side	6	0.332	40
AI-AI	Non-starter-side	7	0.332	40
AI-AI	Non-starter-side	8	0.337	40
AI-AI	Starter-side	2	0.316	40
AI-AI	Starter-side	3	0.327	40
AI-AI	Starter-side	4	0.333	40
AI-AI	Starter-side	5	0.336	40
AI-AI	Starter-side	6	0.342	40
AI-AI	Starter-side	7	0.339	40
AI-AI	Starter-side	8	0.335	40



11 Supplemental: Interleaved vs Speaker View



12 Summary

Primary cross-condition result: per-story $A_{\text{semexp}} = |\text{AUC}_{\text{slot 1}} - \text{AUC}_{\text{slot 2}}|$ from trapezoidal integration of the distance(k) curve over the intersected k range. Tested with ANOVA + Kruskal and pairwise BH-adjusted Wilcoxon. A secondary log-distance-slope asymmetry is reported for dynamic-shape continuity with the paper.

Robustness layers added:

- bootstrap 95% CIs on per-condition mean of A_{semexp}
- pairwise Cliff's δ effect sizes
- planned single-df contrast HA vs (HH+AA)/2

Supplemental views retained:

- speaker-level slot trajectories (descriptive, both slots side-by-side without signed contrast)
- interleaved sequence model and condition descriptives
- within-condition role curves (descriptive only; HA Human/AI inferential view lives in analysis/human-ai/exploration.Rmd)
- interleaved vs speaker-specific side-by-side view

Deferred to a follow-up pass:

- starter-mechanism analysis (signed, starter-normalized AUC asymmetry across all conditions)

Analysis completed: 2026-05-01 23:17:09.710909